

JANAKISRIJA MADDUKURI

Data Engineer / Data Analyst

PROFESSIONAL SUMMARY

Data Engineer with over 5 years of experience in crafting scalable data solutions on AWS and Azure platforms. Demonstrates proficiency in building robust data pipelines for real-time analytics and fraud detection using technologies like Spark and Glue, while collaborating effectively with cross-functional teams to enhance data-driven decision-making. Adept at translating complex business requirements into efficient data models, driving innovation and operational excellence.

EMPLOYMENT HISTORY

DATA ENGINEER II <i>Amazon Web Services</i>	Oct 2024 - Present <i>Seattle</i>
◆ Led the development of end-to-end data pipelines using AWS Glue, EMR, and Spark for processing streaming and batch data, significantly improving data freshness and enabling near-real-time fraud detection. ◆ Built and optimized scalable data pipelines for real-time fraud detection and prevention utilizing AWS services such as Kinesis, Lambda, EMR, and Glue, enabling faster anomaly detection and response. ◆ Automated data workflows with Python, AWS Lambda, and CloudFormation to orchestrate data processing tasks, reducing manual effort and minimizing errors. ◆ Managed large-scale data ingestion and processing across S3, DynamoDB, and Redshift, supporting comprehensive fraud analytics and reporting. ◆ Designed secure, high-availability data storage solutions using S3, RDS, and DynamoDB, ensuring compliance and data integrity through IAM policies and VPC configurations. ◆ Developed custom data transformation jobs leveraging EMR Spark and Hive, improving detection accuracy and reducing false positives. ◆ Configured and monitored CloudWatch, SNS, and SQS for real-time alerts, ensuring high operational health and rapid response to suspicious activities. ◆ Utilized AWS Athena for ad-hoc querying and complex analytics, rapidly generating insights critical for fraud analysis. ◆ Collaborated with Data Science and Machine Learning teams by providing processed data and feature engineering support for fraud detection models. ◆ Created infrastructure as code with CloudFormation, deploying consistent and scalable data processing environments. ◆ Enhanced system performance and cost efficiency by tuning resource utilization and implementing automated scaling policies for EC2, EMR, and ECS clusters. ◆ Developed monitoring dashboards with CloudWatch and Kibana to visualize key performance metrics and detect anomalies proactively. ◆ Designed and implemented complex data pipelines, ensuring rapid processing and analysis, which led to improved fraud detection accuracy. ◆ Partnered with security teams to develop robust data protection strategies, significantly reducing vulnerabilities and enhancing compliance. ◆ Evaluated data flow efficiency, identifying bottlenecks and optimizing workflows, resulting in noticeable improvements in processing speed.	
AWS DATA ENGINEER <i>Cognizant Technology Services</i>	Sep 2022 - Sep 2024
◆ Developed Spark-based applications using Python and Apache Spark to process data from various RDBMS and streaming sources, improving processing efficiency and accuracy. ◆ Created scalable ETL pipelines on AWS EMR for data extraction, cleansing, transformation, and aggregation across diverse formats (JSON, CSV, XML), supporting real-time analytics.	

- ◆ Engineered ETL workflows on AWS EMR to significantly enhance data processing speed and reliability for analytics and reporting.
- ◆ Built and optimized data pipelines utilizing AWS Redshift stored procedures, facilitating seamless data movement and transformation aligned with business needs.
- ◆ Collaborated with cross-functional teams to translate business requirements into optimized stored procedures and data models within AWS Redshift, enabling efficient data analysis.
- ◆ Automated monitoring and performance tuning using AWS CloudWatch and Redshift Query Performance Insights, effectively identifying and addressing bottlenecks.
- ◆ Managed secure data storage and transfer solutions across AWS S3, EC2, and VPC, ensuring compliance, high availability, and resource optimization in public and private subnets.
- ◆ Configured and maintained AWS services such as EC2, S3, Elastic Load Balancer, Security Groups, and NACL to support scalable and secure data architectures.
- ◆ Developed Power BI dashboards and visualizations, empowering stakeholders with actionable business insights.
- ◆ Implemented real-time data processing using Spark Streaming APIs, supporting timely insights and anomaly detection.
- ◆ Automated deployment and CI/CD processes by establishing Jenkins pipelines, improving code integration and release efficiency.
- ◆ Migrated on-premises Hadoop applications to AWS EMR, ensuring scalability and operational continuity.
- ◆ Leveraged Python libraries (pandas, NumPy, SQL Alchemy) for advanced data analysis, transformation, and integration workflows.
- ◆ Designed and maintained logical and physical data models to support robust reporting solutions across diverse business units.
- ◆ Applied data validation and cleansing techniques using SQL, enhancing data quality and consistency throughout processing pipelines.

DATA ENGINEER

Jun 2019 - Jul 2021

McKesson Corporation

- ◆ Streamlined ETL processes using Azure tools, boosting data accessibility and accuracy.
- ◆ Migrated Hadoop applications to Spark, enhancing performance and efficiency.
- ◆ Constructed robust pipelines in Azure Data Factory, improving data integration.
- ◆ Developed Python scripts for ETL workflows, ensuring seamless data processing.
- ◆ Configured real-time data streaming from Apache Kafka, increasing data flow efficiency.
- ◆ Developed automated workflows using Python and Airflow, enhancing operational efficiency and reducing manual intervention in data management tasks.
- ◆ Maintained comprehensive documentation of data transformations and schema changes, ensuring clarity and compliance with corporate standards.
- ◆ Utilized Spark Data Frames for data cleansing and transformation, resulting in noticeable gains in data quality and reliability for reporting.
- ◆ Fostered teamwork by coordinating with cross-functional teams to integrate diverse data sources, strengthening overall project outcomes and stakeholder satisfaction.
- ◆ Refined data processing pipelines, leading to measurable improvements in data accuracy and reporting speed across multiple departments.
- ◆ Implemented efficient ETL frameworks using Azure, significantly enhancing data accessibility for stakeholders and reducing retrieval times.

DATA ENGINEER INTERN

Nov 2018 - May 2019

Luthra Group

India

- ◆ Optimized SQL data models, enhancing query performance and data accuracy.
- ◆ Built database objects and automated ETL processes, improving data handling efficiency.
- ◆ Implemented Spark with PySpark, accelerating data processing speed.
- ◆ Developed CI/CD pipelines, streamlining data workflow integration.

- ◆ Applied Agile methods, enhancing team collaboration and task management.
- ◆ Developed and optimized complex SQL scripts for data extraction, leading to enhanced data accuracy and streamlined reporting processes.
- ◆ Conducted rigorous performance tuning of data models, yielding noticeable improvements in query execution times and overall system efficiency.
- ◆ Fostered teamwork through Agile methodologies, enhancing communication among team members and facilitating timely project deliverables.

EDUCATION

MASTER OF SCIENCE IN BUSINESS ANALYTICS
Kent State University at Kent

Aug 2021 - Aug 2022
Kent, OH

BACHELOR'S IN COMPUTER SCIENCE
JNTUK

AP, India

SKILLS

Hadoop, Databricks, Hive, Apache Spark, EC2, EMR, S3, Lambda, HDFS, Glue, Athena, DynamoDB, RedShift, Azure Data Factory, Azure Data Lake, Azure SQL Database, Azure Synapse, Azure SQL Data Warehouse, Sqoop, Flume, Snowflake, Kinesis, RDS, VPC, CloudWatch, Python, R, SQL, Python Scripting, Pyspark, Shell Scripting, AWS, Azure, MySQL, SQL Server, Oracle, NoSQL, GIT, Bit Bucket, Jenkins, GitHub, JIRA, Postman, Git, Excel, Tableau, Power BI, Matplotlib, Agile Methodologies, Project Management, Data Warehousing, Business Intelligence.

LINKS

LinkedIn: www.linkedin.com, GitHub Portfolio: github.com.

ACADEMIC PROJECTS

Wanderlust Bundle

Developed Models for Trail segmentation and recommendations using K-Means clustering and decision trees on All-Trails data, implemented in R.

News Classifier

Created an interactive web app that classifies news articles into four categories (Business, Entertainment, Technology, Health) using NLP and Machine Learning, achieving 96% accuracy.

ABC Wireless Inc

Built predictive models (logistic regression and decision trees) to determine customer churn based on various factors.

Loan Acceptance Classification Model

Used KNN to create models that predict whether potential customers would accept a personal loan offer.

Cognos Audit Dashboard

Designed a Power BI dashboard for a university project to report on past and current usage of Cognos reports for the leadership team.

CERTIFICATIONS

AWS Solutions Architect – Associate